



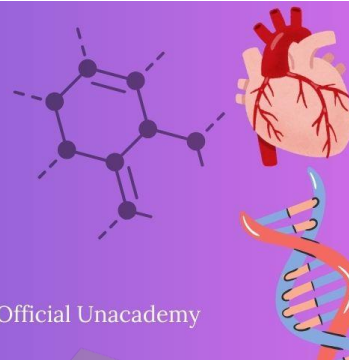
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LAST 10 YEARS

NEET PYQ



Chapter 10: Cell Cycle and Cell Division

1. What is the main function of the spindle fibers during mitosis? [NEET - 2025]

- (1) To separate the chromosomes
- (2) To synthesize new DNA
- (3) To repair damaged DNA
- (4) To regulate cell growth

2. Recombination between homologous chromosomes is completed by the end of [Re-NEET - 2024]

- (1) Diakinesis
- (2) Zygotene
- (3) Diplotene
- (4) Pachytene

3. Match List-I with List-II. [Re-NEET - 2024]

List-I [Event]	List-II [Stage of Prophase-I (Meiosis-I)]
A. Chiasmata formation	I. Pachytene
B. Crossing over	II. Diakinesis
C. Synaptonemal complex formation	III. Diplotene
D. Terminalisation of chiasmata	IV. Zygotene

Choose the correct answer from the options given below :

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-I, C-III, D-IV
- (3) A-III, B-I, C-II, D-IV
- (4) A-II, B-III, C-IV, D-I

4. Match List-I with List-II. [Re-NEET - 2024]

List-I	List-II
A. Cells are metabolically active and proliferate	I. G2 phase
B. DNA replication takes place	II. G1 phase
C. Proteins are synthesised	III. G0 phase
D. Quiescent stage with metabolically active cells	IV. S phase
(1) A-IV, B-II, C-III, D-I	(2) A-I, B-III, C-IV, D-II
(3) A-I, B-I, C-III, D-IV	(4) A-II, B-IV, C-I, D-III

5. Spindle fibers attach to kinetochores of chromosomes during

[NEET - 2024]

- (1) Prophase
- (2) Metaphase
- (3) Anaphase
- (4) Telophase

6. Given below are two statements:

[NEET - 2024]

Statement I : Chromosomes become gradually visible under light microscope during leptotene stage.

Statement II : The beginning of diplotene stage is recognized by dissolution of synaptonemal complex.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

7. Match List I with List II :

[NEET - 2024]

List I	List II
A. Diakinesis	I. Synaptonemal complex formation
B. Pachytene	II. Completion of terminalisation of chiasmata
C. Zygotene	III. Chromosomes look like thin threads
D. Leptotene	IV. Appearance of recombination nodules

- (1) A-IV, B-II, C-III, D-I
- (2) A-I, B-II, C-IV, D-III
- (3) A-II, B-IV, C-I, D-II
- (4) A-IV, B-III, C-II, D-I

8. Following are the stages of cell division :

[NEET - 2024]

- A. Gap 2 phase
- B. Cytokinesis
- C. Synthesis phase
- D. Karyokinesis
- E. Gap 1 phase

Choose the correct sequence of stages from the options given below :

- (1) C-E-D-A-B
- (2) E-B-D-A-C
- (3) B-D-E-A-C
- (4) E-C-A-D-B

9. Doubling of the number of chromosomes can be achieved by disrupting mitotic cell division soon after
(1) Prophase [NEET – 2023-MANIPUR]
(2) Metaphase
(3) Anaphase
(4) Telophase
10. The dissolution of synaptonemal complex occurs during [NEET – 2023-MANIPUR]
(1) Diakinesis
(2) Leptotene
(3) Pachytene
(4) Diplotene
11. During which stages of mitosis and meiosis, respectively does the centromere of each chromosome split? [NEET – 2023-MANIPUR]
(1) Telophase, Anaphase I
(2) Anaphase, Anaphase II
(3) Metaphase, Metaphase II
(4) Prophase, Telophase I
12. Select correct sequence of substages of Prophase-I of meiotic division: [NEET – 2023-MANIPUR]
(A) Zygotene
(B) Pachytene
(C) Diakinesis
(D) Leptotene
(E) Diplotene
Choose the correct answer from the options given below:
(1) (D), (A), (B), (E), (C)
(2) (A), (D), (B), (C), (E)
(3) (D), (B), (A), (E), (C)
(4) (A), (B), (D), (E), (C)
13. Which one of the following is the quiescent stage of cell cycle ? [NEET – 2023-MANIPUR]
(1) G1
(2) G0
(3) M
(4) G2

14. Which of the following stages of meiosis involves division of centromere? [NEET – 2023]

- (1) Metaphase I
- (2) Metaphase II
- (3) Anaphase II
- (4) Telophase

15. The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis? [NEET – 2023]

- (1) Zygotene
- (2) Pachytene
- (3) Diplotene
- (4) Diakinesis

16. Among eukaryotes, replication of DNA takes place in : [NEET – 2023]

- (1) M phase
- (2) S phase
- (3) G1 phase
- (4) G2 phase

17. Match List I with List II : [NEET – 2023]

List I	List II
A. M Phase	I. Proteins are synthesized
B. G2 Phase	II. Inactive phase
C. Quiescent stage	III. Interval between mitosis and initiation of DNA replication
D. G1 Phase	IV. Equational division

Choose the correct answer from the options given below :

- (1) A-III, B-II, C-IV, D-I
- (2) A-IV, B-II, C-I, D-III
- (3) A-IV, B-I, C-II, D-III
- (4) A-II, B-IV, C-I, D-III

18. Given below are two statements: [NEET – 2023]

Statement I : During G₀ phase of cell cycle, the cell is metabolically inactive.

Statement II : The centrosome undergoes duplication during S phase of interphase.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

19. Select the correct statements.

[NEET – 2023]

- A. Tetrad formation is seen during Leptotene.
- B. During Anaphase, the centromeres split and chromatids separate.
- C. Terminalization takes place during Pachytene.
- D. Nucleolus, Golgi complex and ER are reformed during Telophase.
- E. Crossing over takes place between sister chromatids of homologous chromosome.

Choose the correct answer from the options given below:

- (1) A and C only
- (2) B and D only
- (3) A, C and E only
- (4) B and E only

20. The appearance of recombination nodules on homologous chromosomes during meiosis characterizes:

[NEET – 2022]

- (1) Bivalent
- (2) Sites at which crossing over occurs
- (3) Terminalization
- (4) Synaptonemal complex

21. Which one of the following never occurs during mitotic cell division ?

[NEET – 2022]

- (1) Movement of centrioles towards opposite poles
- (2) Pairing of homologous chromosomes
- (3) Coiling and condensation of the chromatids
- (4) Spindle fibres attach to kinetochores of chromosomes

22. Select the incorrect statement with reference to mitosis:

[NEET – 2022]

- (1) Spindle fibres attach to centromere of chromosomes.
- (2) Chromosomes decondense at telophase.
- (3) Splitting of centromere occurs at anaphase.
- (4) All the chromosomes lie at the equator at metaphase.

23. Regarding Meiosis, which of the statements is incorrect ?

[NEET – 2022]

- (1) DNA replication occurs in S phase of Meiosis-II
- (2) Pairing of homologous chromosomes and recombination occurs in Meiosis-I
- (3) Four haploid cells are formed at the end of Meiosis-II
- (4) There are two stages in Meiosis, Meiosis-I and II

24. Which stage of meiosis can last for months or years in the oocytes of some vertebrates ? [Re-NEET – 2022]

- (1) Leptotene
- (2) Pachytene
- (3) Diplotene
- (4) Diakinesis

25. In meiosis, crossing over and exchange of genetic material between homologous chromosomes are catalyzed by the enzyme. [Re-NEET – 2022]

- (1) Phosphorylase
- (2) Recombinase
- (3) Transferase
- (4) Polymerase

26. Identify the correct sequence of events during Prophase I of meiosis : [Re-NEET – 2022]

- (a) Synapsis of homologous chromosomes
- (b) Chromosomes become gradually visible under microscope
- (c) Crossing over between non-sister chromatids of homologous chromosomes
- (d) Terminalisation of chiasmata
- (e) Dissolution of synaptonemal complex

Choose the correct answer from the options given below :

- (1) (a), (b), (c), (d), (e)
- (2) (b), (c), (d), (e), (a)
- (3) (b), (a), (c), (e), (d)
- (4) (a), (c), (d), (e), (b)

27. Bivalent or Tetrad formation is a characteristic feature observed during : [Re-NEET – 2022]

- (1) Synaptonemal complex in zygotene stage
- (2) Chiasmata in Diplotene stage
- (3) Synaptonemal complex in Pachytene stage
- (4) Chiasmata in zygotene stage

28. With respect to metaphase, which of the following statements is incorrect ? [Re-NEET – 2022]

- (1) Complete disintegration of nuclear envelope takes place
- (2) Chromosomes are highly condensed
- (3) Metaphase chromosomes are made up of four sister chromatids held together by centromere
- (4) Chromosomes lie at the equator of the cell

29. Which of the following stages of meiosis involves division of centromere ? [NEET – 2021]
- (1) Metaphase I
 - (2) Metaphase II
 - (3) Anaphase II
 - (4) Telophase II
30. The fruit fly has 8 chromosomes ($2n$) in each cell. During interphase of Mitosis if the number of chromosomes at G1 phase is 8, what would be the number of chromosomes after S phase ? [NEET – 2021]
- (1) 8
 - (2) 16
 - (3) 4
 - (4) 32
31. The centriole undergoes duplication during: [NEET – 2021]
- (1) S-phase
 - (2) Prophase
 - (3) Metaphase
 - (4) G2 phase
32. Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature? [NEET – 2021]
- (1) Leptotene
 - (2) Zygotene
 - (3) Diakinesis
 - (4) Pachytene
33. Identify the correct statement with regard to G1 phase (Gap 1) of interphase. [NEET-2020]
- (1) Nuclear Division takes place.
 - (2) DNA synthesis or replication takes place.
 - (3) Reorganisation of all cell components takes place.
 - (4) Cell is metabolically active, grows but does not replicate its DNA.
34. Match the following with respect to meiosis: [NEET-2020]
- | | |
|----------------|---------------------|
| (a) Zygotene | (i) Terminalization |
| (b) Pachytene | (ii) Chiasmata |
| (c) Diplotene | (iii) Crossing over |
| (d) Diakinesis | (iv) Synapsis |
- (a) (b) (c) (d)
- (1) (ii) (iv) (iii) (i)
 - (2) (iii) (iv) (i) (ii)
 - (3) (iv) (iii) (ii) (i)
 - (4) (i) (ii) (iv) (iii)

35. Some dividing cells exit the cell cycle and enter vegetative inactive stage. This is called quiescent stage (G₀).

This process occurs at the end of :

[NEET-2020]

- (1) G₂ phase
- (2) M phase
- (3) G₁ phase
- (4) S phase

36. Dissolution of the synaptonemal complex occurs during :

[NEET-2020]

- (1) Leptotene
- (2) Pachytene
- (3) Zygotene
- (4) Diplotene

37. Attachment of spindle fibres to kinetochores of chromosomes becomes evident in : [NEET-2020(Covid-19)]

- (1) Anaphase
- (2) Telophase
- (3) Prophase
- (4) Metaphase

38. In a mitotic cycle, the correct sequence of phases is

[NEET-2020(Covid-19)]

- (1) S, G₁, G₂, M
- (2) G₁, S, G₂, M
- (3) M, G₁, G₂, S
- (4) G₁, G₂, S, M

39. During Meiosis-I, in which stage synapsis takes place ?

[NEET-2020(Covid-19)]

- | | |
|---------------|---------------|
| (1) Pachytene | (2) Zygotene |
| (3) Diplotene | (4) Leptotene |

40. Match the following events that occur in their respective phases of cell cycle and select the correct option :

- | | | |
|--------------------------|---|-----------------------|
| (a) G ₁ phase | (i) Cell grows and organelle duplication | [NEET-2020(Covid-19)] |
| (b) S phase | (ii) DNA replication and chromosome duplication | |
| (c) G ₂ phase | (iii) Cytoplasmic growth | |
| (d) Metaphase in M-phase | (iv) Alignment of chromosomes | |

- (1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

41. The correct sequence of phases of cell cycle is :

[NEET-2019]

- (1) $M \rightarrow G_1 \rightarrow G_2 \rightarrow S$
- (2) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$
- (3) $S \rightarrow G_1 \rightarrow G_2 \rightarrow M$
- (4) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$

42. Cells in G_0 phase:

[NEET-2019]

- (1) exit the cell cycle
- (2) enter the cell cycle
- (3) suspend the cell cycle
- (4) terminate the cell cycle

43. After meiosis-I, the resultant daughter cells have:-

[NEET-2019(Odisha)]

- (1) Same amount of DNA as in the parent cell in S phase
- (2) Twice the amount of DNA in comparison to haploid gamete.
- (3) Same amount of DNA in comparison to haploid gamete
- (4) Four times the amount of DNA in comparison to haploid gamete

44. Crossing over takes place between which chromatids and in which stage of the cell cycle ?

[NEET-2019(Odisha)]

- (1) Non-sister chromatids of non-homologous chromosomes at Zygotene stage of prophase I.
- (2) Non-sister chromatids of homologous chromosomes at Pachytene stage of prophase I.
- (3) Non-sister chromatids of homologous chromosomes at Zygotene stage of prophase I.
- (4) Non-sister chromatids of non homologous chromosomes at Pachytene stage of prophase I.

45. The stage during which separation of the paired homologous chromosomes begins is

[NEET-2018]

- (1) Pachytene
- (2) Diplotene
- (3) Diakinesis
- (4) Zygotene

46. Which of the following options gives the correct sequence of events during mitosis ?

[NEET-2017]

- (1) Condensation $\rightarrow$ nuclear membrane disassembly $\rightarrow$ arrangement at equator $\rightarrow$ centromere division $\rightarrow$ segregation $\rightarrow$ telophase
- (2) Condensation $\rightarrow$ crossing over $\rightarrow$ nuclear membrane disassembly $\rightarrow$ segregation $\rightarrow$ telophase
- (3) Condensation $\rightarrow$ arrangement at equator $\rightarrow$ centromere division $\rightarrow$ segregation $\rightarrow$ telophase
- (4) Condensation $\rightarrow$ nuclear membrane disassembly $\rightarrow$ crossing over $\rightarrow$ segregation $\rightarrow$ telophase

47. Anaphase Promoting Complex (APC) is a protein degradation machinery necessary for proper mitosis of animal cells. If APC is defective in a human cell, which of the following is expected to occur ? [NEET-2017]

- (1) Chromosomes will be fragmented
- (2) Chromosomes will not segregate
- (3) Recombination of chromosome arms will occur
- (4) Chromosomes will not condense

48. Spindle fibres attach on to :-

[NEET- I 2016]

- (1) Telomere of the chromosome
- (2) Kinetochore of the chromosome
- (3) Centromere of the chromosome
- (4) Kinetosome of the chromosome

49. In meiosis crossing over is initiated at :

[NEET- I 2016]

- (1) Pachytene
- (2) Leptotene
- (3) Zygotene
- (4) Diplotene

50. During cell growth, DNA synthesis takes place in:-

[NEET- II 2016]

- (1) G2 phase
- (2) M phase
- (3) S phase
- (4) G1 phase

51. When cell has stalled DNA replication fork, which checkpoint should be predominantly activated?

[NEET- II 2016]

- (1) M
- (2) Both G2/M and M
- (3) G1/S
- (4) G2/M

52. Match the stages of meiosis in Column-I to their characteristic features in Column-II and select the correct option using the codes given below :

[NEET- II 2016]

Column-I

Column-II

- | | |
|----------------|---|
| a. Pachytene | i. Pairing of homologous chromosomes |
| b. Metaphase-I | ii. Terminalization of chiasmata |
| c. Diakinesis | iii. Crossing over takes place |
| d. Zygotene | iv. Chromosomes align at equatorial plate |

a b c d

- (1) ii iv iii i
- (2) iv iii ii i
- (3) iii iv ii i
- (4) i iv ii iii

ANSWER KEY

1. (1)
2. (4)
3. (1)
4. (4)
5. (2)
6. (1)
7. (3)
8. (4)
9. (2)
10. (4)
11. (2)
12. (1)
13. (2)
14. (3)
15. (2)
16. (2)
17. (3)
18. (4)
19. (2)
20. (2)
21. (2)
22. (1)
23. (1)
24. (3)
25. (2)
26. (3)
27. (1)
28. (3)
29. (3)
30. (1)
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34. (3)
35. (3)
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39. (2)
40. (4)
41. (4)

- 42. (3)
- 43. (2)
- 44. (2)
- 45. (2)
- 46. (1)
- 47. (2)
- 48. (2)
- 49. (1)
- 50. (3)
- 51. (4)
- 52. (3)

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